CHEM 110 Spring 2026 S3

Assignment 5

Due on Monday (February 9th) 6 pm

(To get the full points, you need to write down the solving or calculation process)

1. The second-order decomposition of NO_2 has a rate constant of 0.255 $M^{-1}s^{-1}$. How much NO_2 decomposes in 4.00 s if the initial concentration of NO_2 is 1.33 mole/L?

2. The first-order reaction $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ has a half-life of 8.75 hours at 593 K. How long will it take for the concentration of SO_2Cl_2 to fall to 16.5% of its initial value?

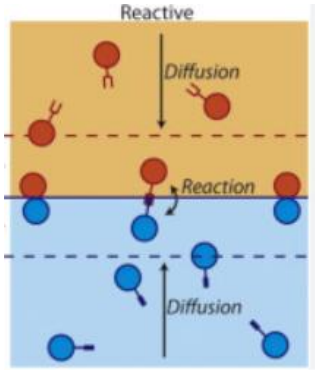
3. Identify the type(s) of the catalyst: (A) heterogeneous or (B) homogeneous. Explain your reason briefly.

(1) Use Iron to catalyze the reaction between nitrogen and hydrogen gas

(2) Use graphene to catalyze the redox reaction of H_2O_2 and potassium permanganate in water.

(3) Use HCl solution to catalyze the hydrolysis of water-soluble esters.

(4) Use micelles to catalyze the interfacial reactions between water and oil phases.



4. The following first-order reaction has a rate constant of 0.76 s^{-1} at 1000K. How long will it take for the concentration of N_2O to decrease to 12% of its initial concentration?

$$2\ N_2O\ (g) \rightarrow 2\ N_2\ (g) + O_2\ (g)$$

5. A first-order reaction is measured to have a rate constant of $3.61 \times 10^{-15}\ s^{-1}$ at 298 K and a rate constant of $8.66 \times 10^{-7}\ s^{-1}$ at 425 K. What is the activation energy for this reaction?

6. Determine the rate constant and reaction order by plotting the right mathematical form of concentration vs time. Attach the Excel plot with the trendline and the fitting equation below.

A→ 2D

Time (s)	[A] (M)
0	0.05000
60	0.01376
120	0.00798
180	0.00562
240	0.00433
300	0.00353
360	0.00298
420	0.00257
480	0.00227
540	0.00202
600	0.00183
660	0.00167
720	0.00153
780	0.00142
840	0.00132

7. Hydrogen iodide decomposes at 800K via a second-order process to produce hydrogen and iodine according to the following reaction $2HI(g) \rightarrow H_2(g) + I_2(g)$ At 800K, it takes 142 seconds for the initial concentration of HI to decrease from 6.75×10^{-2} M to 3.50×10^{-2} M. What is the rate constant for the reaction at this temperature?

8. True OR False. Briefly explain the reason if it is false.

(1) _____ Rate constants have the same unit for all reactions.

(2) _____ The following reaction has a molecularity of 2.

$$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g) \quad Rate=k[NO][O_3]$$

(3) _____ The half-life of a first-order reaction is dependent on the initial concentration of reactant.

(4) _____ The addition of a homogeneous catalyst does not change the activation energy of a given reaction while a heterogeneous one does.

(5) _____The reaction $2\ N_2O \rightarrow 2\ N_2 + O_2$ has the rate relationships:

$$\text{Rate} = \frac{1}{2}\frac{\Delta[N_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t} = -\frac{1}{2}\frac{\Delta[N_2O]}{\Delta t}$$

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(6) _____ The order of reaction can only be determined experimentally.